Set-Membership Estimation Based Distributed Cooperative Localization of Multiple UAVs

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***Abstract*—This paper investigates the distributed cooperative localization problem for mobile multi-UAV systems using set-membership estimation (SME). Unlike existing SME approaches relying on prior-known anchors or beacons, we consider a fully autonomous scenario where UAVs need to collaboratively derive guaranteed positional sets using only absolute measurements and relative range measurements from neighboring agents. To handle this problem, we first propose a law-of-total-range method to rigorously quantify the estimated sets from relative measurements, which fits the general form of relative measurements between pairwise UAVs. Based on this new method, a novel distributed SME framework is developed, where each UAV can obtain the guaranteed set containing its true state with only local measurements and communications. In addition, a distributed localization algorithm based on extended constrained zonotopes (ECZ) is designed with a proposed framework. Simulation results and field experiments verify that our proposed method efficiently improves the cooperative localization accuracy and outperforms the linearization-based extended SME method.**

***Index Terms*—Set-membership estimation, multi-UAV system, cooperative localization.**

I. Introduction

In RECENT years, the cooperative localization problem of multi-agent systems has received considerable research attention [1], [2], [3]. Various estimation methods have been widely used in the cooperative localization problem.

For observer-based methods, a distributed self-localization algorithm was developed for multi-agent systems by using relative position measurements in [4]. The results were extended to the distributed localization problem of sources with bearing angle measurements in [5]. In [6], distributed consensus filters were developed for a set of targets to utilize relative velocity and distance measurements to estimate the relative positions between UAVs. For stochastic methods, distributed localization algorithms are often designed based on Bayesian theory [7], [8]. When the process and measurement noise comply with the Gaussian distribution, the distributed Kalman filter has been proposed to deal with the problem, which has attracted significant attention. In [9] and [10], the authors designed distributed Kalman filters for a multi-agent system with linear relative measurements, and transformed the covariance update into a sparse-constrained matrix trace minimization problem. In [11], the author designed a distributed Kalman filter for the cooperative localization problem with integrated absolute and relative measurements. In [12], a new adaptive extended Kalman filter is proposed to solve the problem of unknown noise covariance matrices inherent in the cooperative localization of autonomous underwater vehicles. In [13], nonlinear relative measurements have been considered, where a recursive quadratic filter has been proposed to compute the exact value of the variance explicitly. To deal with energy-bounded noise, authors have proposed an H_∞-filter-based method in [14] to realize cooperative localization. In [15], a robust approach is proposed to obtain optimal multi-robot relative observations using the Metric-based Iterative Closest Point algorithm, which enables robots to utilize the surrounding environment information directly instead of placing a localization mark on the robots. It is worth mentioning that the aforementioned methods have limitations when facing unknown but bounded (UBB) noises. To the best of our knowledge, the unknown-input observer (UIO) [16] and the set-membership estimation (SME) are two effective ways to deal with the UBB noise.

In this paper, we focus on the guaranteed method, i.e., set-membership estimation (SME), to address the cooperative localization problem in multi-agent systems. Based on our current information, only a few articles have focused on studying the SME based cooperative localization problem. In [17], the author considered a multi-agent system with both absolute and relative position measurements. A constrained zonotope [18] based algorithm was developed to enable distributed cooperative localization. In [19], the indoor localization problem based on the received signal strength indicator (RSSI) is considered. The author proposed a distributed set-membership estimation method in which the state of the localization system is uncertain and constrained by a given ellipsoid. In [20], Silvestre investigated the set-membership-based localization problem of relative range and bearing measurement with a fixed beacon. A new tool of set description, w.r.t., the constrained convex generator (CCG) is proposed as a generalization of the constrained zonotope and applied in the localization problem to derive the estimated guaranteed set. Qin et al. considered the same localization problem [21], and proposed a moving-horizon extended constrained zonotopic algorithm to accelerate calculations. References [22] and [23] studied the SME-based localization problem using GPS

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and UWB range measurements, which is similar to this work. The authors introduced a novel Iterated Extended Set Membership Filter (IESMF) that generates ellipsoidal regions guaranteed to contain the position of another aircraft. However, these two articles focused on relative localization between agents, whereas we are more interested in the problem where each UAV estimates its own position.

It should be pointed out that existing works on SME-based localization rely on prior knowledge of anchors or beacons, which may be unrealistic in practical applications, particularly in outdoor environments. Moreover, most works handle nonlinear relative measurements by linearizing simultaneous equations. This is because the state coupling caused by relative measurements is difficult to decouple and form distributed frameworks.

Thus, in this article, we study the SME-based distributed cooperative localization problem of a multi-UAV system, where each UAV aims to generate a guaranteed set that contains its own position. The main contributions of this paper are summarized as follows:

- By analyzing uncertain variables, we accurately describe the uncertain range determined by general relative measurements between two agents, which forms the basis for establishing the distributed SME framework.
- A distributed SME framework is developed that can handle kinematic scenarios without prior knowledge of anchors. With the proposed framework, each UAV can obtain the guaranteed set that contains its true state using only local measurements and communications.
- An extended constrained zonotopic distributed localization algorithm is designed based on the proposed SME framework. Simulations verify that the designed algorithm can effectively improve accuracy with cooperative information from relative measurements, and outperforms the linearization-based extended SME method in [23].

The rest of the paper is organized as follows: Section II presents the system model and problem description. Main results are provided in Section III, with the distributed SMFing framework analyzed in Section III-B and the constrained zonotopic algorithm designed in Section IV. Section V presents simulation results that corroborate the effectiveness of our methods. Finally, Section VI provides the concluding remarks of the paper.

Notations: For an uncertain variable $\mathbf{x}$, its realization is defined by $\mathbf{x}(\omega) = x$. The set of all the realizations of an uncertain variable is called the uncertain range: $[\![\mathbf{x}]\!] = \{x(\omega) : \omega \in \Omega\}$ [24]. $\mathbb{N}_0$ is the set of natural number. $\mathbb{R}^n$ denotes the n-dimensional Euclidean space. For arbitrary sets S_1, S_2, their Minkowski sum is denoted by $S_1 \oplus S_2 = \{s_1 + s_2 : s_1 \in S_1,\ s_2 \in S_2\}$. The interval hull of a bounded set $S \ni s = (s^{(1)}, \ldots, s^{(n)})$ is $\overline{\mathrm{IH}}(S) = \prod_{i=1}^{n} [\underline{s}^{(i)}, \overline{s}^{(i)}]$, where $\underline{s}^{(i)} = \inf_{s^{(i)}}\{s \in S\}$ and $\overline{s}^{(i)} = \sup_{s^{(i)}}\{s \in S\}$.

II. System Model and Problem Description

Consider a system composed of N UAVs, each with the discrete-time second-order model

$$x_{i,k+1} = A_i x_{i,k} + B_i(u_{i,k} + w_{i,k}), \tag{1}$$

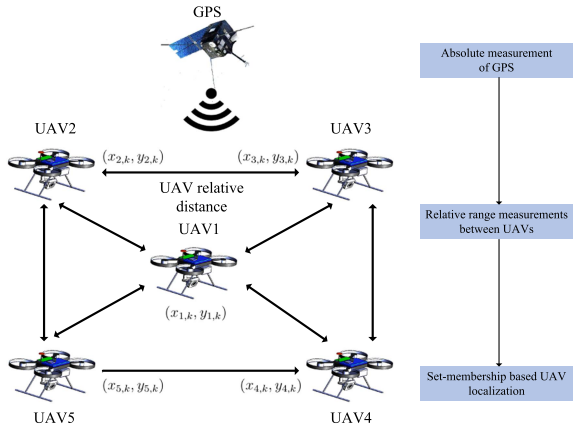


Fig. 1. Graphical Illustration.

where the state $x_{i,k} = [p_{i,k}^T, \nu_{i,k}^T]^T$ includes the position $p_{i,k}$ and velocity $\nu_{i,k}$ of UAV i at time instant k, and

$$A_i = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} \frac{T^2}{2} \\ T \end{bmatrix}. \tag{2}$$

$u_{i,k}$ denotes the input of UAV i at step k, which is assumed as a known signal to i itself in this article. $w_{i,k} \in [\![\mathbf{w}_{i,k}]\!] \subseteq \mathbb{R}^{n_i}$ is the process noise.

In this work, two types of measurements in the cooperation problems [11], [13], i.e., absolute measurements and relative measurements are considered.

Absolute measurement: each UAV i takes an observation on its state directly, i.e., with the following form:

$$y_{i,k} = C_i x_{i,k} + v_{i,k}, \tag{3}$$

$y_{i,k} \in \mathbb{R}^{m_i}$ is the measurement at time instant k; $C_i = I_2$ is the measurement matrix; $v_{i,k}$ is the bounded measurement noise, i.e., $v_{i,k} \in [\![\mathbf{v}_{i,k}]\!]$.

Relative measurements: each UAV utilizes the Ultra Wide Band (UWB) sensor to take relative range measurements with its neighbors:

$$y_{i,j,k}^r = \|p_{i,k} - p_{j,k}\|_2 + r_{i,j,k},\ j \in \mathcal{N}_i, \tag{4}$$

where $\|\cdot\|_2$ is the 2-norm of a vector, and $y_{i,j,k}^r \in \mathbb{R}^{p_i}$ is the relative distance between agent i and its in-neighborhood agent j at time instant k; $r_{i,j,k}$ is the bounded range measurement noise, with $r_{i,j,k} \in [\![\mathbf{r}_{i,j,k}]\!] = [\underline{r}_{i,j,k}, \overline{r}_{i,j,k}]$.

Unless otherwise stated, we assume the unrelatedness of the initial state of each UAV and the uncertain variables of noise at each time are assumed to guarantee the non-stochastic Hidden Markov Model [25], i.e., $\forall k \in \mathbb{N}_0$, $\mathbf{w}_{i,0:k}, \mathbf{v}_{i,0:k}, \mathbf{r}_{i,j,0:k}$, $\mathbf{x}_{i,0}$, $\forall i, j = 1, \ldots, N$ are unrelated.

In this paper, we study the cooperative localization problem where UAVs utilize the absolute measurement and relative measurements with neighbors to realize cooperative localization (see Fig. 1). Specially, the aim of each UAV is to utilize all the local information $y_{i,k}$ and $y_{i,j,k}^r$, $j \in \mathcal{N}_i$ up to k to derive an estimated set $S_{i,k}$, such that $x_{i,k} \in S_{i,k}, \forall k \in \mathbb{N}_0$.

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III. Uncertain Range Analysis and Distributed SME Framework

In this section, the uncertain range corresponding to general relative measurements is first derived Section III-A as a basis. Based on this, a distributed SME framework is established in Section III-B.

A. Uncertain Range Determined by Relative Measurements

Consider UAV i and its neighbor $j \in \mathcal{N}_i$, with the following relative measurement equation,

$$z_{i,j,k} = g_{i,j}(x_{i,k}, x_{j,k}) + r_{i,j,k}, \tag{5}$$

when the uncertain range of UAV j, $S_{j,k}$ is given, we focus on how to obtain an uncertain range $S_{i,k}$ such that $x_{i,k} \in S_{i,k}$, and have the following lemma.

Lemma 1: For any given estimated set $S_{j,k} \ni x_{j,k}$, the estimated set $S^r_{i,k}$ derived by the relative measurement (5) is presented by

$$S^r_{i,j,k} = g^{-1}_{i,j,1}(\{z_{i,j,k}\} \oplus [\![-\mathbf{r}_{i,j,k}]\!], S_{j,k}). \tag{6}$$

where $g^{-1}_{i,j}$ denotes the the preimage of $\{z_{i,j,k} - r^0_{i,j,k}\}$ under mapping $g_{i,j}(\cdot, x^0_{j,k})$,

$$g^{-1}_{i,j}\left(\{z_{i,j,k} - r^0_{i,j,k}\}, x^0_{j,k}\right) = \{x_{i,k} : g_{i,j}(x_{i,k}, x^0_{j,k}) + r^0_{i,j,k} = z_{i,j,k}\}. \tag{7}$$

Proof: See Appendix A. ■

For the relative range measurement (4), Lemma 1 can be specialized as the following proposition.

Proposition 1: The estimated set $S^r_{i,k}$ derived by the relative range measurement (4) satisfies:

$$S^r_{i,j,k} = S_{j,k} \oplus Y^r_{i,j,k}, \tag{8}$$

$Y^r_{i,j,k}$ is a geometrical ring defined by

$$Y^r_{i,j,k} = \{s : \underline{R}_{i,j,k} \le \|s\|_2 \le \overline{R}_{i,j,k}\}, \tag{9}$$

where $\overline{R}_{i,j,k} = y^r_{i,j,k} - \overline{r}_{i,j,k}$ and $\underline{R}_{i,j,k} = y^r_{i,j,k} - \underline{r}_{i,j,k}$.

Proof: See Appendix B. ■

B. Distributed SME Framework

For the multi-UAVs system with dynamics (1) and measurements (3), (4), the distributed SME process can be expressed by the following iterative framework, i.e., Algorithm 1.

A line-by-line explanation of Algorithm 1 is provided below. Line 1 initializes the algorithm. Lines 2-7 give the estimation process at each $k \in \mathbb{N}_0$. Line 3 predicts the prior estimated set $S_{i,k}$ when the posterior estimated set of the last instant $S_{i,k-1}$ is given. Line 4 describes the process of measurements and communications. The information of the estimated set $\overline{S}_{j,k} \cap S^y_{j,k}$ can be transmitted from j to i through communications. Line 5 updates the posterior estimated set $S_{i,k}$, which is the intersection of prior set $\overline{S}_{i,k}$ and the measurement set $S^y_{i,k}$, $S^r_{i,j,k}$, $j \in \mathcal{N}_i$.

Theorem 1: When the initial state satisfies $x_{i,0} \in S_{i,0}$, $\forall i \in \mathcal{V}$, the posterior range $S_{i,k}$ of each UAV generated by Algorithm 1 contains the true state $x_{i,k}$ for any $k \in \mathbb{N}_0$.

Algorithm 1: Distributed SME Framework.

1: **Initialization**. The initial uncertain range of each UAV i is settled as $S_{i,0}$.
2: **loop**
3: **Prediction**. The predicted prior range $\overline{S}_{i,k}$ is given by

$$\overline{S}_{i,k} = A_i(S_{i,k-1}) \oplus \{B_i u_{i,k-1}\} \oplus B_i[\![\mathbf{w}_{i,k-1}]\!], \tag{10}$$

where $S_{i,k-1}$ is the posterior range at step $k-1$.
4: **Measurement and communications**. Each UAV takes an absolute measurement $y_{i,k}$. UAV i takes relative range measurements $y^r_{i,j,k}$, and receives information from neighbor UAV j.
5: **Update**. The posterior estimated set $S_{i,k}$ is presented by

$$S_{i,k} = S^{abs}_{i,k} \cap \bigcap_{j \in \mathcal{N}_i} [S^{abs}_{j,k} \oplus Y^r_{i,j,k}], \tag{11}$$

where $S^{abs}_{i,k} = \overline{S}_{i,k} \cap S^y_{i,k}$, and $S^y_{i,k} = \{x_{i,k} : y_{i,k} = C_i x_{i,k} + v_{i,k},\ v_{i,k} \in [\![\mathbf{v}_{i,k}]\!]\}$.
6: $k = k + 1$;
7: **end loop**

Proof: See Appendix C. ■

IV. Extended Constrained Zonotope Based Localization Algorithm Design

Since the calculations under general set descriptions are challenging, geometric figures such as ellipsoids and zonotopes are often used to outer bound $S_{i,k}$. Generally speaking, the constrained zonotopic algorithm has less over-estimation conservativeness than ellipsoid-based algorithms, which implies its better accuracy [26]. Thus, in this section, we design a distributed cooperative localization algorithm based on Algorithm 1, utilizing the extended constrained zonotope. Specifically, our goal is to find extended constrained zonotopes $\mathcal{Z}_{i,k}$ such that $\mathcal{Z}_{i,k} \supseteq S_{i,k}$. The definition of an extended constrained zonotope is presented as follows.

Definition 1: (Extended Constrained Zonotope [27]) An extended constrained zonotope $\mathcal{Z} \subseteq \mathbb{R}^n$ is described by a quintuple $(G, c, A, b, h) \in \mathbb{R}^{n \times n_g} \times \mathbb{R}^n \times \mathbb{R}^{n_c \times n_g} \times \mathbb{R}^{n_c} \times [0, \infty]^{n_g}$:

$$Z(G, c, A, b, h) = \left\{ G\xi + c : A\xi = b, \xi \in \prod_{j=1}^{n_g} \left[-h^{(j)}, h^{(j)}\right] \right\}, \tag{12}$$

where $h^{(j)}$ is the jth component of h[1].

Compared to the polytope or zonotope, the extended constrained zonotope is sufficient to describe arbitrary convex

[1]When $h^{(j)} = 1$, $j = 1, \ldots, n_g$, Definition 1 becomes the classical constrained zonotope in [18].

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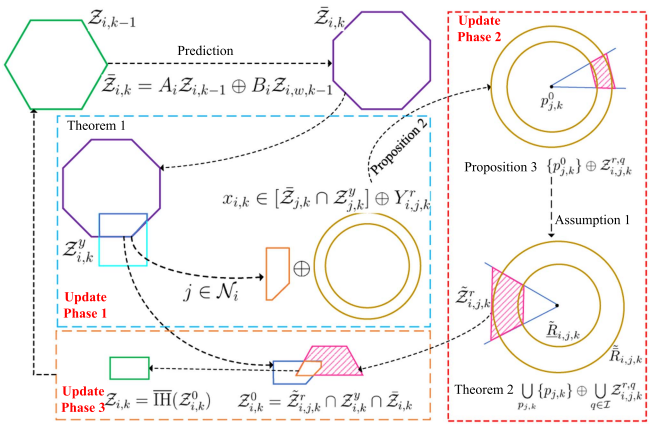


Fig. 2. The green hexagon denotes the posterior extended constrained zonotope of UAV i at step $k-1$, $\mathcal{Z}_{i,k-1}$; the purple octagon denotes the prior set $\overline{\mathcal{Z}}_{i,k}$; $\mathcal{Z}_{i,k}^{abs}$ is presented by the blue irregular polygon. According to Proposition 2, the calculation of $\mathcal{Z}_{j,k}^{abs} \oplus Y_{i,j,k}^{r}$ is divided into two steps, which are detailed in Proposition 3 and Theorem 2, respectively.

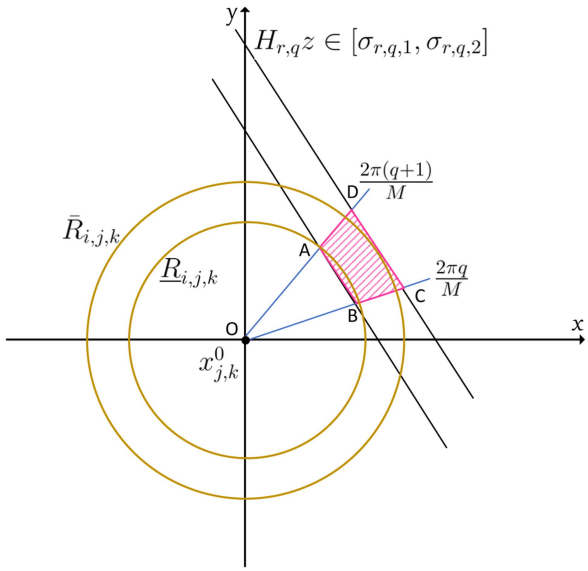


Fig. 3. The diagram of proof of Proposition 3.

polygons, and permits the computation of exact projections, intersections, and Minkowski sums.

Lemma 2: For every $Z = (G_z, c_z, A_z, b_z, h_z) \subset \mathbb{R}^n$, $W = (G_w, c_w, A_w, b_w, h_w) \subset \mathbb{R}^n$, $Y = (G_y, c_y, A_y, b_y, h_y) \subset \mathbb{R}^k$ and $T \in \mathbb{R}^{k\times n}$, the three identities hold:

$$TZ = (TG_z, Tc_z, A_z, b_z, h_z), \tag{13}$$

$$Z \oplus W = \left([G_z\ G_w], c_z + c_w, \begin{bmatrix} A_z & 0 \\ 0 & A_w \end{bmatrix}, \begin{bmatrix} b_z \\ b_w \end{bmatrix}, \begin{bmatrix} h_z \\ h_w \end{bmatrix}\right), \tag{14}$$

$$Z \cap_{\mathbf{T}} Y = \left([G_z\ 0], c_z, \begin{bmatrix} A_z & 0 \\ 0 & A_y \\ TG_z & -G_y \end{bmatrix}, \begin{bmatrix} b_z \\ b_y \\ c_y - Tc_z \end{bmatrix}, \begin{bmatrix} h_z \\ h_y \end{bmatrix}\right). \tag{15}$$

Lemma 2 implies the extended constrained zonotope is closed under linear transformation, Minkowski sum and intersection operations. As a result, all items in Line 5 of Algorithm 1, except $S_{j,k}^{abs} \oplus Y_{i,j,k}^{r}$, can be directly obtained with Lemma 2. Therefore, in the following analysis, we will focus on the calculation of $S_{j,k}^{abs} \oplus Y_{i,j,k}^{r}$, which is inspired by the following proposition.

Proposition 2: Let $R = \bigcup_{q\in\mathcal{I}} R^q$, the following equation holds

$$S_1 \oplus R = \bigcup_{s\in S_1} \bigcup_{q\in\mathcal{I}} (\{s\} \oplus R^q). \tag{16}$$

Proof: This conclusion comes directly from the commutativity of set union and Minkowski sum, i.e., $(S_1 \oplus \bigcup_{q\in\mathcal{I}} R^q) = \bigcup_{q\in\mathcal{I}} (S_1 \oplus R^q)$. ■

According to Proposition 2, the derivation of $S_{j,k}^{a} \oplus Y_{i,j,k}^{r}$ is performed in two steps, as shown in Fig. 2. In the first step, we select a fixed point $p_{j,k}^0 \in S_{j,k} \cap S_{j,k}^{y}$, and a segmented sector $Y_{i,j,k}^{r,q}$ is outer bounded by an extended constrained zonotope $\mathcal{Z}_{i,j,k}^{r,q}$, as presented in Proposition 3. Building on this, Theorem 2 calculates the union of $\bigcup_{p_{j,k}} \{p_{j,k}\} \oplus \bigcup_{q\in\mathcal{I}} Y_{i,j,k}^{r,q}$.

For the fixed point $p_{j,k}^0 \in \mathcal{Z}_{j,k}^{abs}$, we divide the ring $Y_{i,j,k}^{r}$ into M parts. Each part $Y_{i,j,k}^{r,q}$ has a minimum angle value of $\frac{2\pi q}{M}$ and a maximum angle value $\frac{2\pi(q+1)}{M}$, $q = 1, \ldots, M$. We have the following proposition (see Fig. 3).

Proposition 3: The sector, w.r.t., $\{p_{j,k}^0\} \oplus Y_{i,j,k}^{r,q}$ is outer bounded by the following extended constrained zonotope

$$\mathcal{Z}_{i,j,k}^{r,q} = \Bigg\{ z : z = [G_{r,q}\ 0] \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} + c_{r,q},$$

$$\begin{bmatrix} H_{r,q}G_{r,q} & (\sigma_{r,q,1} - \sigma_{r,q,2}) \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \sigma_{r,q,1} - H_{r,q}c_{r,q},$$

$$\begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} \in h_{r,q} \times [0,1] \Bigg\}, \tag{17}$$

where

$$G_{r,q} = \begin{bmatrix} \cos\left(\frac{2\pi(q+1)}{M}\right) & \cos\left(\frac{2\pi(q)}{M}\right) \\ \sin\left(\frac{2\pi(q+1)}{M}\right) & \sin\left(\frac{2\pi(q)}{M}\right) \end{bmatrix},$$

$$c_{r,q} = p_{j,k}^0,$$

$$H_{r,q} = \begin{bmatrix} \cos\left(\frac{\pi(2q+1)}{M}\right) & \sin\left(\frac{\pi(2q+1)}{M}\right) \end{bmatrix},$$

$$\sigma_{r,q,1} = \underline{R}_{i,j,k} \cos\left(\frac{2\pi(q+1)}{M}\right) + H_{r,q}c_{r,q},$$

$$\sigma_{r,q,2} = \overline{R}_{i,j,k} + H_{r,q}c_{r,q},$$

$$h_{r,q} = \left[0, \frac{\overline{R}_{i,j,k}}{\cos\left(\frac{\pi q}{M}\right)}\right] \times \left[0, \frac{\overline{R}_{i,j,k}}{\cos\left(\frac{\pi q}{M}\right)}\right]. \tag{18}$$

Proof: See Appendix D. ■

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Remark 1: Based on Proposition 3, each set $\mathcal{Z}_{i,j,k}^{1,r}, \ldots, \mathcal{Z}_{i,j,k}^{M,r}$ can be intersected with $\mathcal{Z}_{i,k}^{abs}$, and discarded if $\mathcal{Z}_{i,j,k}^{q,r} \cap \mathcal{Z}_{i,k}^{abs} = \emptyset$ to reduce computation burden.

To generalize the one-point-scene to the union $\bigcup_{p_{j,k} \in \mathcal{Z}_{j,k}^{abs}}$, we make the following assumption.

Assumption 1: The diameter of $\mathcal{Z}_{j,k}^{abs}$ satisfies $d(\mathcal{Z}_{i,k}^{abs}) \ll \underline{R}_{i,j,k} < \delta(p_{j,k}^0, \mathcal{Z}_{i,k}^{abs}), \forall i \in \mathcal{V}, j \in \mathcal{N}_i$, where $\delta(p_{j,k}^0, \mathcal{Z}_{i,k}^{abs}) = \inf_{p_i \in \mathcal{Z}_{i,k}^{abs}} \|p_{j,k}^0 - p_i\|_2$ represents the distance between $p_{j,k}^0$ and set $\mathcal{Z}_{i,k}^{abs}$.

Remark 2: Assumption 1 implies the position of UAV j is far from UAV i. This assumption is reasonable, as the relative distance between UAVs is typically much larger than the uncertainty introduced by noise.

Based on Proposition 3 and Assumption 1, an extended constrained zonotope can be derived to outer bound $(\mathcal{Z}_{j,k}^{abs} \oplus Y_{i,j,k}^{r}) \cap \mathcal{Z}_{i,k}^{abs}$.

Theorem 2: Under Assumption 1, there holds $(\mathcal{Z}_{j,k}^{abs} \oplus Y_{i,j,k}^{r}) \cap \mathcal{Z}_{i,k}^{abs} \subseteq \widetilde{\mathcal{Z}}_{i,j,k}^{r}$, where

$$\widetilde{\mathcal{Z}}_{i,j,k}^{r} = \left\{ z : z = [\widetilde{G}_r \; 0] \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} + \widetilde{c}_r, \right.$$
$$\left. \begin{bmatrix} \widetilde{H}_r \widetilde{G}_r & (\widetilde{\sigma}_{r,1} - \widetilde{\sigma}_{r,2}) \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \widetilde{\sigma}_{r,1} - \widetilde{H}_r \widetilde{c}_r, \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} \in \widetilde{h}_r \right\}, \tag{19}$$

with parameters defined as follows,

$$\widetilde{G}_r = \begin{bmatrix} \cos\left(\frac{2\pi(\widetilde{q}_2+1)}{M}\right) & \cos\left(\frac{2\pi\widetilde{q}_1}{M}\right) \\ \sin\left(\frac{2\pi(\widetilde{q}_2+1)}{M}\right) & \sin\left(\frac{2\pi\widetilde{q}_1}{M}\right) \end{bmatrix};$$
$$\widetilde{c}_r = \forall p_{j,k}^* \in \mathcal{Z}_{j,k}^{abs},$$
$$\widetilde{\sigma}_{r,1} = \left(\underline{\widetilde{R}}_{i,j,k}\right) \cos\left(\frac{(\widetilde{q}_2 - \widetilde{q}_1 + 1)\pi}{M}\right) + \widetilde{H}_r \widetilde{c}_r,$$
$$\widetilde{\sigma}_{r,2} = \overline{\widetilde{R}}_{i,j,k} + \widetilde{H}_r \widetilde{c}_r,$$
$$\widetilde{H}_r = \left[\cos\left(\frac{\pi(\widetilde{q}_1+\widetilde{q}_2+1)}{M}\right) \quad \sin\left(\frac{\pi(\widetilde{q}_1+\widetilde{q}_2+1)}{M}\right)\right],$$
$$\widetilde{h}_r = \left[0, \frac{\overline{\widetilde{R}}_{i,j,k}}{\cos\left(\frac{(\widetilde{q}_2-\widetilde{q}_1+1)\pi}{M}\right)}\right] \times \left[0, \frac{\overline{\widetilde{R}}_{i,j,k}}{\cos\left(\frac{(\widetilde{q}_2-\widetilde{q}_1+1)\pi}{M}\right)}\right] \times [0,1]$$
$$\overline{\widetilde{R}}_{i,j,k} = d(\mathcal{Z}_{j,k}^{abs}) + \overline{R}_{i,j,k},$$
$$\underline{\widetilde{R}}_{i,j,k} = \underline{R}_{i,j,k} - d(\mathcal{Z}_{j,k}^{abs})$$
$$\widetilde{q}_1 = \inf_q \mathcal{Z}_{i,j,k}^{q,r} \cap \mathcal{Z}_{i,k}^{abs} \neq \emptyset,$$
$$\widetilde{q}_2 = \sup_q \mathcal{Z}_{i,j,k}^{q,r} \cap \mathcal{Z}_{i,k}^{abs} \neq \emptyset. \tag{20}$$

Proof: See Appendix E. ■

To summarize, the process of extended constrained zonotope-based distributed cooperative localization is shown as in Algorithm 2.

Algorithm 2: Extended Constrained Zonotope Based Distributed Cooperative Localization.

1: **Initialization**. Set the initial extended constrained zonotope of each UAV i, $\mathcal{Z}_{i,0}$.
2: **loop**
3: **Prediction**. The predicted extended constrained zonotope $\overline{\mathcal{Z}}_{i,k}$ is given by
$$\overline{\mathcal{Z}}_{i,k} = A_i \mathcal{Z}_{i,k-1} \oplus \{B_i u_{i,k-1}\} \oplus B_i \mathcal{Z}_{w_i}, \tag{21}$$
where $\mathcal{Z}_{i,k-1}$ is the posterior range at step $k-1$.
4: **Measurement and communications**. Each UAV takes absolute measurement $y_{i,k}$, and calculates $\mathcal{Z}_{i,k}^{abs} = \overline{\mathcal{Z}}_{i,k} \cap \mathcal{Z}_{i,k}^{y}$. UAV i takes relative range measurements $y_{i,j,k}^r$, and receives information from neighbor UAV j.
5: **Update**. The posterior extended constrained zonotope $\mathcal{Z}_{i,k}$ is calculated as in Lines 6-15.
6: Let $p_{j,k}^* = c_{j,k}^{abs}$, the diameter of $\mathcal{Z}_{j,k}^{abs}$ is given by $d(\mathcal{Z}_{j,k}^{abs}) = \sup_{\|\xi\| \leq 1} \|G_{j,k}^{abs} \xi\|_\infty \leq \|G_{j,k}^{abs}\|_\infty$.
7: The equivalent relative range measurement set is given by the ring $\widetilde{Y}_{i,j,k}^{r} = \{s : \underline{\widetilde{R}}_{i,j,k} \leq \|s\|_2 \leq \overline{\widetilde{R}}_{i,j,k}\}$, where $\underline{\widetilde{R}}_{i,j,k} = y_{i,j,k}^r - \overline{r}_{i,j,k} - d(\mathcal{Z}_{j,k}^{abs})$ and $\overline{\widetilde{R}}_{i,j,k} = y_{i,j,k}^r - \underline{r}_{i,j,k} + d(\mathcal{Z}_{j,k}^{abs})$.
8: Segment $\widetilde{Y}_{i,j,k}^{r}$ into M parts. Each part is outer bounded by the extended constrained zonotope $\widetilde{Z}_{i,j,k}^{q,r}$ as in (18).
9: **loop**
10: Calculate $\widetilde{Z}_{i,j,k}^{q,r} \cap \mathcal{Z}_{i,k}^{abs}$, $q \in [1, M]$.
11: $q = q + 1$.
12: **end loop**
13: Let $q_1 = \inf_q \widetilde{Z}_{i,j,k}^{q,r} \cap \mathcal{Z}_{i,k}^{abs} \neq \emptyset$ and $q_2 = \sup_q \widetilde{Z}_{i,j,k}^{q,r} \cap \mathcal{Z}_{i,k}^{abs} \neq \emptyset$. Then the extended constrained zonotope $\widetilde{\mathcal{Z}}_{i,j,k}^{r}$ given by the relative range measurement $y_{i,j,k}^r$ can be represented as (19) and (20).
14: Calculate $\mathcal{Z}_{i,k}^0 = \overline{\mathcal{Z}}_{i,k} \cap \widetilde{\mathcal{Z}}_{i,j,k}^{r} \cap \mathcal{Z}_{i,k}^{y}$ with (13).
15: Calculate the interval hull as the posterior estimated set of UAV i, i.e., $\mathcal{Z}_{i,k} = \overline{\mathrm{IH}}(\mathcal{Z}_{i,k}^0)$.
16: $k = k + 1$;
17: **end loop**

Remark 3: To illustrate the calculation of $\widetilde{q}_1$ and $\widetilde{q}_2$, we consider the example in Fig. 4. The geometrical ring is devided into M parts. Each part is labeled with $1, \ldots, M$ in an anti-clockwise order, and can be outer bounded by an extended constrained zonotope $\mathcal{Z}_{i,j,k}^{r,q}$ as in (17) and (18). Notice that the intersections $\mathcal{Z}_{i,j,k}^{r,1} \cap \mathcal{Z}_{i,k}^{abs}$ and $\mathcal{Z}_{i,j,k}^{r,2} \cap \mathcal{Z}_{i,k}^{abs}$ are not empty,

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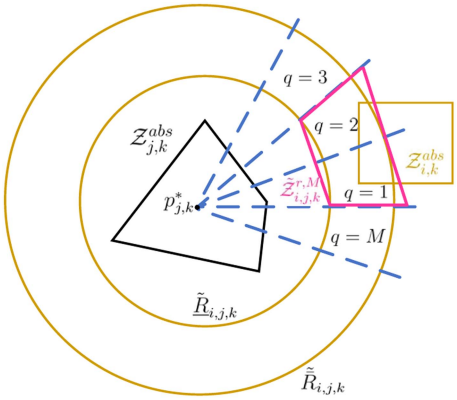


Fig. 4. The diagram of the calculation of $\widetilde{q}_1$ and $\widetilde{q}_2$. Only the first and second segmented sector have non-empty intersections with $\mathcal{Z}^{abs}_{i,k}$, thus the other sectors can be discarded. We have $\widetilde{q}_1 = 1$ and $\widetilde{q}_2 = 2$.

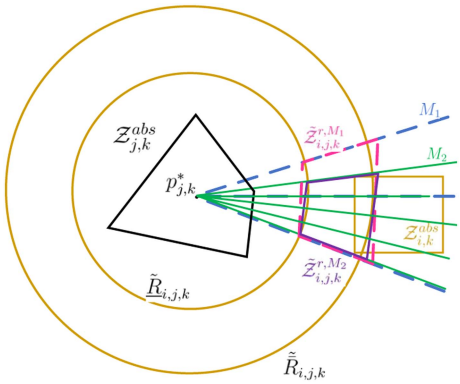


Fig. 5. Illustration of different estimation results when choosing different M ($M_1 < M_2$). When $M = M_1$, the posterior estimated set generated by Algorithm 2 is the intersection of $\widetilde{\mathcal{Z}}^{r,M_1}_{i,j,k}$ (the pink dash trapezoid) with $\mathcal{Z}^{abs}_{i,k}$. When M increases to M_2, the posterior estimated set is the intersection of $\widetilde{\mathcal{Z}}^{r,M_2}_{i,j,k}$ (the purple trapezoid) with $\mathcal{Z}^{abs}_{i,k}$. It can be observed that the latter has less over-estimation error, which implies the improvement of estimation accuracy as M increases.

from the definition of $\widetilde{q}_1$ and $\widetilde{q}_2$ (see (20)), we have $\widetilde{q}_1 = 1$ and $\widetilde{q}_2 = 2$.

Remark 4: The choice of M has a trade-off on estimation accuracy and computational efficiency.

On one hand, the estimation accuracy increases as M increases. Algorithm 2 is to utilize the union of several extended constrained zonotope to cover the intersection of $\mathcal{Z}^{abs}_{i,k}$ and the geometrical ring. Thus, the over-estimation error is unavoidable since some part of the cones are redundant. When M increases, each cone gets thinner, which implies the area of redundant part decreases. As shown in Fig. 5, when the ring is segmented into M_1 parts. The cone generated by each part is represented by the blue dash line. Notice the two cones in Fig. 5 has non-empty intersections with $\mathcal{Z}^{abs}_{i,k}$, thus the posterior extended constrained zonotope is presented by the intersection of $\widetilde{\mathcal{Z}}^{r,M_1}_{i,j,k}$ (the pink dash trapezoid in Fig. 5 and $\mathcal{Z}^{abs}_{i,k}$. When the segmented number increases to M_2, the cone generated by each part is represented by the green line. The posterior estimated set is presented by the intersection of $\widetilde{\mathcal{Z}}^{r,M_2}_{i,j,k}$ (the purple trapezoid in Fig. 5 and $\mathcal{Z}^{abs}_{i,k}$.

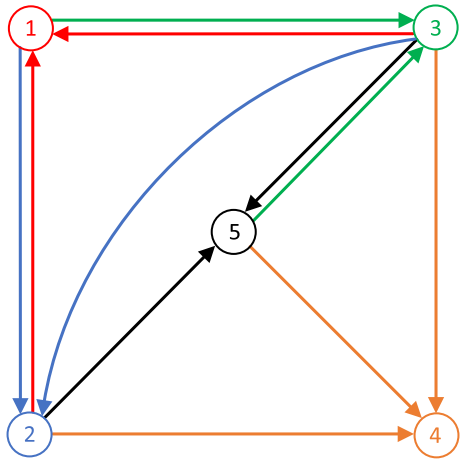


Fig. 6. System measurement topology.

Apparently, the purple trapezoid introduces less over-estimation error than the pink dash trapezoid, which implies better estimation accuracy for $M_2 > M_1$.

On the other hand, Algorithm 2 needs to judge whether each segmented part has empty intersections with $\mathcal{Z}^{abs}_{i,k}$, which implies its contains M times set-intersection calculations and empty set judgements. Thus, its computational efficiency decreases as M increases.

Remark 5: When UAV i takes relative bearing and range measurements simultaneously with its neighbor j, where the relative bearing measurement equation is as follows,

$$y^b_{i,j,k} = \text{ang}(x_{i,k} - x_{j,k}) + r^b_{i,j,k}, \tag{22}$$

the posterior extended constrained zonotope of UAV is presented by $\mathcal{Z}^{abs}_{i,k} \cap (\mathcal{Z}^{abs}_{j,k} \oplus \mathcal{Z}^{r,b}_{i,j,k})$, where $\mathcal{Z}^{r,b}_{i,j,k}$ can be obtained by replacing $\frac{2\pi(q+1)}{M}$ and $\frac{2\pi q}{M}$ in (18) with $y^b_{i,j,k} - \overline{r}^b_{i,j,k}$ and $y^b_{i,j,k} - \underline{r}^b_{i,j,k}$, respectively.

V. Simulation and Field Experiment

A. Simulation Example

In this subsection, a 2-D plane localization simulation is used to demonstrate the efficiency of the proposed method. System measurement topology is shown as in Fig. 6. Simulation parameters are set as $T = 0.25$, $C_i = I_4$. The process noise $w_{i,k}$ and absolute measurement noise $v_{i,k}$ are set as

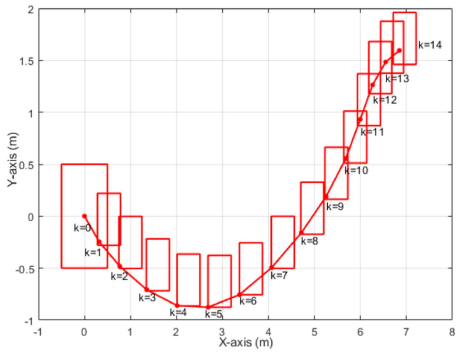
$$w_{1,k} = \frac{1}{4(1+e^{-kT})} \cdot \begin{bmatrix}1\\1\end{bmatrix} + \begin{bmatrix}2\cos(\lfloor\frac{k}{4}\rfloor \cdot \frac{\pi}{2})\\2\sin(\lfloor\frac{k}{4}\rfloor \cdot \frac{\pi}{2})\end{bmatrix},$$

$$w_{2,k} = \frac{1}{4(1+e^{-kT})} \cdot \begin{bmatrix}1\\1\end{bmatrix} + \begin{bmatrix}2\sin(\lfloor\frac{k}{4}\rfloor \cdot \frac{\pi}{2})\\2\cos(\lfloor\frac{k}{4}\rfloor \cdot \frac{\pi}{2})\end{bmatrix},$$

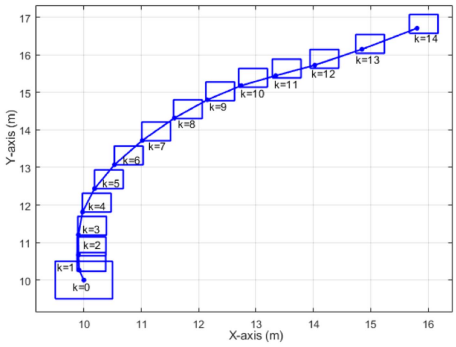
$$w_{3,k} = \frac{1}{4(1+e^{-kT})} \cdot \begin{bmatrix}1\\1\end{bmatrix} + \begin{bmatrix}2\cos((\lfloor\frac{k}{4}\rfloor + 1) \cdot \frac{\pi}{2})\\2\sin((\lfloor\frac{k}{4}\rfloor - 1) \cdot \frac{\pi}{2})\end{bmatrix},$$

$$w_{4,k} = \frac{1}{4(1+e^{-kT})} \cdot \begin{bmatrix}1\\1\end{bmatrix} + \begin{bmatrix}2\sin((\lfloor\frac{k}{4}\rfloor - 1) \cdot \frac{\pi}{2})\\2\cos((\lfloor\frac{k}{4}\rfloor - 1) \cdot \frac{\pi}{2})\end{bmatrix},$$

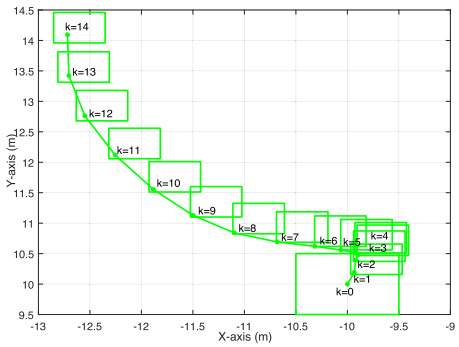
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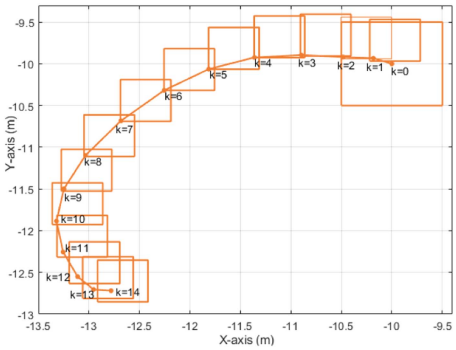
(a) SME based cooperative localization of UAV1



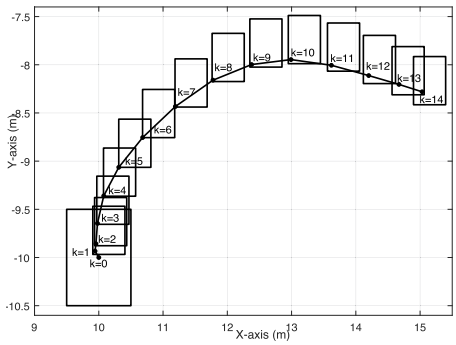
(b) SME based cooperative localization of UAV2



(c) SME based cooperative localization of UAV3



(d) SME based cooperative localization of UAV4



(e) SME based cooperative localization of UAV5

Fig. 7. Simulation to verify the effectiveness of Algorithm 2. The estimated sets (rectangles) generated by Algorithm 2 contain the true positions of UAVs (dots) in each step, which verifies the effectiveness of Algorithm 2.

$$w_{5,k} = \frac{1}{4(1+e^{-kT})} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2\sin(\lfloor \frac{k}{4} \rfloor \cdot \frac{\pi}{2}) \\ 2\cos(\lfloor \frac{k}{4} \rfloor \cdot \frac{\pi}{2}) \end{bmatrix},$$

$$v_{i,k} = 0.25 \cdot \sin\left(0.5kT + \frac{\pi}{4}\right) \cdot \begin{bmatrix} 1 \\ T \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \tag{23}$$

where $\lfloor \cdot \rfloor$ rounds down a real number. One can verify that $w_{i,k} \in [\![\mathbf{w}_{i,k}]\!]$ and $v_{i,k} \in [\![\mathbf{v}_{i,k}]\!]$, where

$$[\![\mathbf{w}_{i,k}]\!] = \{z : z = 2.25 I_2 \xi, \|\xi\|_\infty \leq 1\},$$
$$[\![\mathbf{v}_{i,k}]\!] = \{z : z = 0.25 \cdot I_4 \xi, \|\xi\|_\infty \leq 1\}. \tag{24}$$

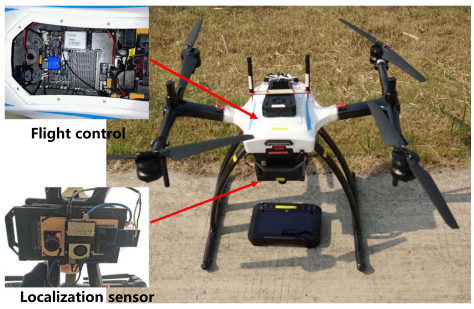
Thus, $w_{i,k}$ and $v_{i,k}$ are bounded noise, with $\|w_{i,k}\| \leq 2.25$ and $\|v_{i,k}\| \leq 1$. The range of the relative measurement noise $r_{i,j,k}$ is set as $[\![\mathbf{r}_{i,j,k}]\!] = [-0.1, 0.1]$. The simulation results for the localization estimation of UAVs are presented in Fig. 7. The line represents the real trajectory, and the true positions at different time steps are marked with dots. The rectangles are the estimated interval hulls generated by Algorithm 2 ($M = 80$). As demonstrated in Fig. 7(a)–(e), the proposed algorithm consistently ensures that the true system state is contained within the interval hull boundaries of the posterior extended constrained zonotopes at all time steps. This indicates the effectiveness of our method.

To quantify the conservativeness of the SME, we adjust the uncertain range of absolute measurement noise as $[\![v_{i,k}]\!] = \{z : z = I_4 \xi, \|\xi\|_\infty \leq 1\}$, and utilize the diameter of the estimated set $d(S)$ as a measure of uncertainty, which represents the maximum estimation error of the estimated set. Specifically, $d(S)$ is defined as:

$$d(S) = \sup_{s_1, s_2 \in S} \|s_1 - s_2\|_\infty. \tag{25}$$

For an interval hull in the 2-D plane, $d(S)$ corresponds to the maximum edge length, which is twice the infinite norm of the generator matrix G. Therefore, we define the following index to

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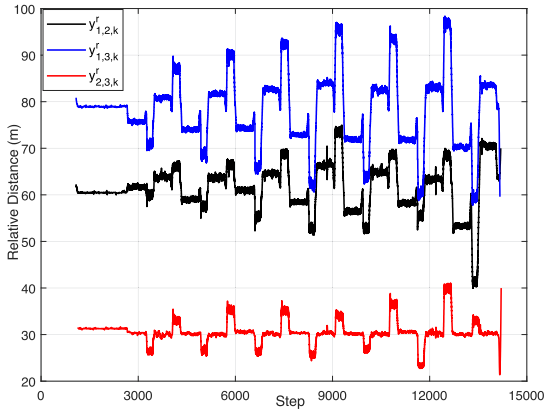
(a) The UAV is equipped with a localization sensor, which is composed of visual, GPS, and UWB sensors.



(b) Flight process of three UAVs.



(c) The desired trajectories of three UAVs.



(d) The relative range measurements between UAVs during the flight process .

Fig. 8. The scenario and equipment for cooperative localization.

TABLE I
CONSERVATIVENESS COMPARISON OF DIFFERENT ALGORITHMS

$[\![\mathbf{r}_{i,j,k}]\!]$	Algorithm	UAV1	UAV2	UAV3	UAV4	UAV5
$[-0.1, 0.1]$	SME in [28]	0.8182	0.8162	0.8155	0.8197	0.8170
	Algorithm 2	**0.7650**	**0.7611**	**0.7618**	**0.7630**	**0.7621**
	ESME in [23]	0.7838	0.7798	0.7852	0.7846	0.7835
	DPF-SME in Appendix F	0.7489	0.7475	0.7482	0.7581	0.7493
$[-0.2, 0.2]$	Algorithm 2	**0.7678**	**0.7632**	**0.7673**	**0.7681**	**0.7672**
	ESME in [23]	0.7883	0.7862	0.7898	0.7909	0.7891
	DPF-SME in Appendix F	0.7602	0.7598	0.7598	0.7611	0.7608
$[-0.5, 0.5]$	Algorithm 2	**0.7720**	**0.7713**	**0.7725**	**0.7734**	**0.7765**
	ESME in [23]	0.7912	0.7910	0.7913	0.7915	0.7924
	DPF-SME in Appendix F	0.7684	0.7692	0.7708	0.7712	0.7731

reflect the average of interval hull diameters:

$$\overline{d}_i = \frac{1}{K}\sum_{k=1}^{K}\|G_{i,k}\|_\infty. \tag{26}$$

Using this index, we compare our method with the non-cooperative localization method [28], the extended set-membership estimation (ESME) [23] and the distributed particle-filter based SME (DPF-SME, see Algorithm 3 in Appendix A). The index $\overline{d}_i$ for different methods is shown in Table I. In Table I, we can observe that the index $\overline{d}_i$ of Algorithm 2 is smaller than that of the non-cooperative SME method. This indicates that our proposed method effectively utilizes cooperation information from relative measurements, thereby improving estimation accuracy.

Furthermore, the $\overline{d}_i$ of Algorithm 2 is lower than the ESME method in reference [23]. This is because in the linearization method, i.e., the ESME, the linearized residuals contains a quadratic item $\triangle_{i,k} = (\widetilde{x}_{i,k} - \overline{x}_{i,k})^T \frac{\partial^2 h(\widetilde{x}_{i,k})}{\partial \widetilde{x}_{i,k}^2}|_{\widetilde{x}_{i,k}=\overline{x}_{i,k}} (\widetilde{x}_{i,k} - \overline{x}_{i,k})$, where $\overline{x}_{i,k}$ is the central point of the prior extended constrained zonotope of UAV i at step k, and $\widetilde{x}_{i,k} \in \overline{\mathcal{Z}}_{i,k}$ is an arbitrary point in $\overline{\mathcal{Z}}_{i,k}$. To guarantee the posterior extended constrained zonotope $\mathcal{Z}_{i,k}$ contain all possible $x_{i,k}$, the range of $\triangle_{i,k}$ should be zero to its maximum, which is quadratic to $\widetilde{x}_{i,k} - \overline{x}_{i,k}$, i.e., the radius of $\overline{\mathcal{Z}}_{i,k}$. On the other hand, the over-estimation of our proposed method mainly comes from the outer approximation of $Y_{i,j,k}^{r,q}$ with $Z_{i,j,k}^{r,q}$. Thus, compared to ESME, our proposed method does not suffer from the over-estimation caused by the quadratic item $\triangle_{i,k}$, and has less conservativeness in estimation.

It should be noticed that the DPF-SME outperforms our proposed algorithm in estimation accuracy. **However**, an important limitation of Algorithm 3 is that it can not guarantee its

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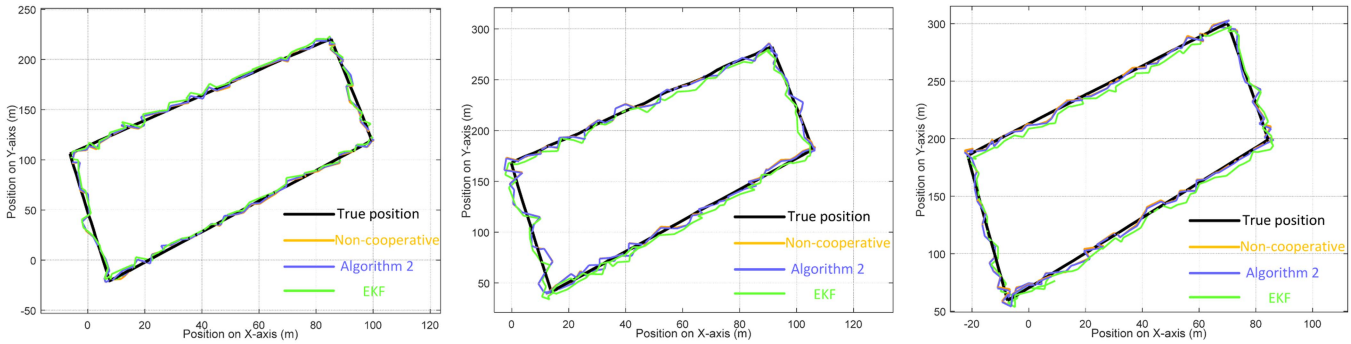


Fig. 9. (a)The true and estimated trajectories of UAV1. (b)The true and estimated trajectories of UAV2. (c)The true and estimated trajectories of UAV3.

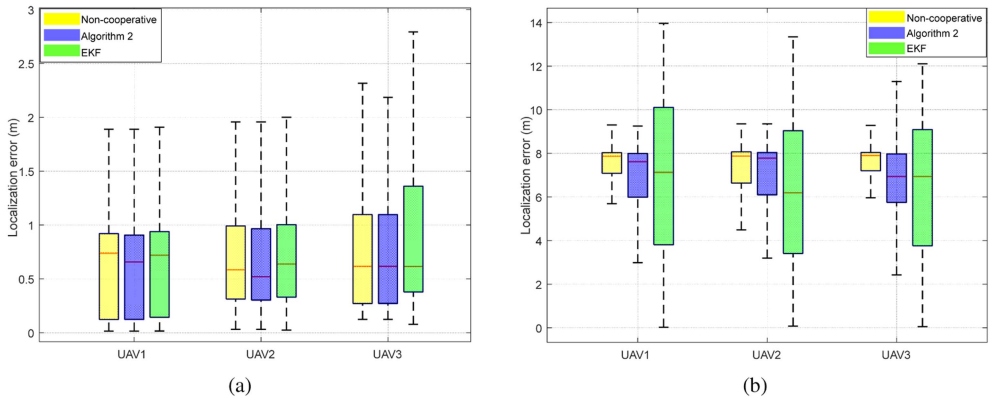


Fig. 10. (a) Estimation error of different methods under GPS undenied scene. (b) Estimation error of different methods under GPS denied scene.

TABLE II
TIME NODE OF FIELD FLIGHT

UAV	Time of taking off	Time of starting GPS denying	Time of Finishing GPS denying	Time of landing
UAV1	15:42:14.052	15:47:03.197	16:17:39.704	16:19:14.087
UAV2	15:42:14.0539	15:47:02.826	16:17:34.382	16:19:15.184
UAV3	15:40:43.460	15:47:06.789	16:17:33.692	16:19:11.997

estimated set contains the true state. Notice that each feasible point generated in Algorithm 3 is an interior point of the true uncertain range, which implies the estimated set generated by Algorithm 3 is still an inner approximation even if the number of generated particles M tends to infinity. Thus, the estimated set generated by Algorithm 3 can not guarantee containing all possible states, which violates the fundamental purpose of SME.

B. *Field Flight Experiment*

To further verify the effectiveness of Algorithm 2, a field experiment was conducted. Three UAVs are tasked with flying along the trajectories shown in Fig. 8(c). The time node of the field flight experiment is shown in Table II. After taking off, each UAV utilizes GPS to take absolute measurements of its own position. The localization accuracy of GPS measurement is $1\,\mathrm{m}$.

After about 5 minutes, the UAVs enter the GPS-denied region and utilize the image-matching method as its absolute measurement with an accuracy of $21\,\mathrm{m}$.

In the whole flight process, UAVs take relative range measurements with the UWB sensor customized by the Sinomap Field Company, with a measurement noise $r_{i,j,k} \in [-10\,\mathrm{m}, -10\,\mathrm{m}]$. The relative distance between UAVs and their neighbors is shown in Fig. 8(d).

Algorithm 2 is utilized to realize measurement fusion, and is considered to be effective if its localization accuracy is better than the image-matching method.

The flight process is divided into two stages. In the first stage, each UAV uses both GPS and a UWB sensor for localization. In the second stage, the GPS signal is denied, and UAVs utilize visual and UWB sensors to localize themselves. The true positions and estimated trajectories of the UAVs are presented in Fig. 9.

Moreover, we calculate the absolute value of pointwise estimation error $e_{i,k} = |\widehat{p}_{i,k} - p_{i,k}|$ of Algorithm 2, and make a comparison with the non-cooperative method, and the EKF method (see Fig. 10). In the box plot Fig. 10, it can be observed that both the mean error and maximum error of the proposed

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Algorithm 2 are lower, indicating that our method also performs well in pointwise estimation.

VI. Conclusion

In this article, we have studied the distributed cooperative localization problem of a multi-UAV system with absolute and relative range measurements based on set-membership estimation (SME). Each kinematic UAV needs to realize SME-based localization with its absolute measurement of itself and relative range measurements of its neighbors. Firstly, to deal with the general form of relative measurements between pairwise UAVs, we have proposed a new method to derive the uncertain range of uncertain variables with the law of total range. Based on this, a novel distributed SME framework has been developed, where each UAV can obtain the guaranteed set containing its true state with only local measurements and communications. Furthermore, we have designed an extended constrained zonotopic localization algorithm as a realization of the proposed distributed SME framework. Finally, several simulation results have corroborated the effectiveness of our proposed method, and our designed algorithm can effectively improve accuracy with cooperative information from relative measurements, and has less conservativeness than the linearization-based estimation method (ESMF and EKF).

For future works, we aim to study the rigid-graph-based cooperative localization problem of dense with SME, which can be applied in dense UAV formations. We also consider the data-driven unknown-input-observer with SME, which can present reliable guaranteed sets and the convergent pointwise estimations simultaneously; when the model contains dynamic level uncertainties, the data-driven unknown-input observer can deal with the model mismatch, and the presented SME method can be applied to realize cooperative localization of each UAV.

Appendix A
Proof of Lemma 1

Proof: For the uncertain variable $\mathbf{x}_{i,k}$, we aim to obtain the conditional marginal range $S^r_{i,k} = [\![\mathbf{x}_{i,k}|z_{i,j,k}]\!]$, which can be derived by the law of total range [21],

$$\begin{aligned}[\![\mathbf{x}_{i,k}|z_{i,j,k}]\!] &= \bigcup_{r_{i,j,k}\in[\![\mathbf{r}_{i,j,k}]\!]} [\![\mathbf{x}_{i,k}|r_{i,j,k}, z_{i,j,k}]\!] \\ &= \bigcup_{r_{i,j,k}\in[\![\mathbf{r}_{i,j,k}]\!]} \bigcup_{x_{j,k}\in S_{j,k}} [\![\mathbf{x}_{i,k}|x_{j,k}, r_{i,j,k}, z_{i,j,k}]\!].\end{aligned} \tag{27}$$

Noting that for each fixed $r^0_{i,j,k} \in [\![\mathbf{r}_{i,j,k}]\!]$ and $x^0_{j,k} \in S_{j,k}$, we have

$$\begin{aligned}&[\![\mathbf{x}_{i,k}|x_{j,k}, r_{i,j,k}, z_{i,j,k}]\!] \\ &\quad = \{x_{i,k} : g_{i,j}(x_{i,k}, x^0_{j,k}) + r^0_{i,j,k} = z_{i,j,k}\}.\end{aligned} \tag{28}$$

The right hand side (RHS) of (28) is the definition of the preimage $\{z_{i,j,k} - r^0_{i,j,k}\}$ under mapping $g_{i,j}(\cdot, x^0_{j,k})$ (see (7)). By substituting (28) into (27), we can obtain

$$\begin{aligned}S^r_{i,j,k} &= \bigcup_{x_{j,k}\in S_{j,k}} \bigcup_{r_{i,j,k}\in[\![\mathbf{r}_{i,j,k}]\!]} g^{-1}_{i,j}(\{z_{i,j,k} - r_{i,j,k}\}, x_{j,k}) \\ &= g^{-1}_{i,j,1}(\{z_{i,j,k}\} \oplus [\![-\mathbf{r}_{i,j,k}]\!], S_{j,k}).\end{aligned} \tag{29}$$

Thus, Lemma 1 has been proved. ■

Appendix B
Proof of Proposition 1

Proof: According to Lemma 1, we have

$$S^r_{i,j,k} = \bigcup_{x_{j,k}\in S_{j,k}} \bigcup_{r_{i,j,k}\in[\![\mathbf{r}_{i,j,k}]\!]} g^{-1}_{i,j}(\{z_{i,j,k} - r^0_{i,j,k}\}, x^0_{j,k}), \tag{30}$$

where

$$\begin{aligned}&g^{-1}_{i,j}(\{z_{i,j,k} - r^0_{i,j,k}\}, x^0_{j,k}) \\ &\quad = \{x_{i,k} : x_{i,k} = x^0_{j,k} + s_0, \|s_0\|_2 = z_{i,j,k} - r^0_{i,j,k}\}\end{aligned} \tag{31}$$

for each given $\underline{r}_{i,j,k} \le r^0_{i,j,k} \le \overline{r}_{i,j,k}$ and $x^0_{j,k} \in S_{j,k}$.

Noting that the union $\bigcup_{r_{i,j,k}\in[\![\mathbf{r}_{i,j,k}]\!]}$ of the RHS of (31) corresponds to the geometric ring (9). Thus, (30) can be written as

$$S^r_{i,j,k} = \{x_{i,k} : x_{i,k} = x_{j,k} + s,\ x_{j,k} \in S_{j,k},\ s \in Y^r_{i,j,k}\}. \tag{32}$$

From the definition of Minkowski sum, Proposition 1 can be proved. ■

Appendix C
Proof of Theorem 1

Proof: When $x_{i,k-1} \in S_{i,k-1}$ holds, from the dynamics (1), we have

$$\begin{aligned}x_{i,k} &= A_i x_{i,k-1} + B_i u_{i,k-1} + B_i w_{i,k-1} \\ &\in A_i S_{i,k-1} \oplus \{B_i u_{i,k-1}\} \oplus B_i [\![\mathbf{w}_{i,k-1}]\!] = \overline{S}_{i,k}.\end{aligned} \tag{33}$$

Moreover, from the definition of $S^y_{i,k}$, we know that $x_{i,k} \in S^y_{i,k}$. Thus, $x_{i,k} \in S^{abs}_{i,k} = \overline{S}_{i,k} \cap S^y_{i,k}$ holds for any $i \in \mathcal{V}$ and $k \in \mathbb{N}_0$. Hence, $x_{j,k} \in S^{abs}_{j,k}$. Together with Proposition 1, we can obtain

$$x_{i,k} \in S^{abs}_{i,k} \cap [S^{abs}_{j,k} \oplus Y^r_{i,j,k}],\ \forall j \in \mathcal{N}_i. \tag{34}$$

Theorem 1 has been proved. ■

Appendix D
Proof of Proposition 3

Proof: According to the geometrical significance, the segmented sector $Y^{r,q}_{i,j,k}$ is outer bounded by the trapezoid ABCD, which is the intersection of triangle $\{z : z \in \triangle\mathrm{COD}\}$ and hyperplane $\{z : H_{r,q} z \in [\sigma_{r,q,1}, \sigma_{r,q,2}]\}$, where $H_{r,q} = \left[\cos\left(\frac{\pi(2q+1)}{M}\right) \quad \sin\left(\frac{\pi(2q+1)}{M}\right)\right]$.

For the triangle $\triangle$COD, each point can be covered by the linear combination of $\frac{\vec{\mathrm{OC}}}{\|\vec{\mathrm{OD}}\|}$ and $\frac{\vec{\mathrm{OD}}}{\|\vec{\mathrm{OD}}\|}$, hence $\triangle$COD can be expressed by the following zonotope:

$$\mathcal{Z}_{COD} = \Big\{ z : z = G_{r,q}\xi_1 + c_{r,q},$$

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Algorithm 3: Particle-Filter Based Distributed SME Algorithm.

Initialization. For each agent i, set the initial prior range $S_{i,0} = [a_{i,x,0}, b_{i,x,0}] \times [a_{i,y,0}, b_{i,y,0}]$. $M = 10000$;{**Comments**: M is the number of random samples for $S_{i,k}$ $(k \in \mathbb{N}_+)$.}

loop

  while $k \in \mathbb{N}$

    for $i = 1 : N$

      for $m = 1 : M$

        for $j \in \mathcal{N}_i$

    Generate random particles $x_{i(j),k-1} \leftarrow U([a_{i(j),x,k-1}, b_{i(j),x,k-1}] \times [a_{i(j),y,k-1}, b_{i(j),y,k-1}])$ and $w_{i(j),k-1} \leftarrow [\![\mathbf{w}_{i(j),k-1}]\!]$; {**Comments**: $x \leftarrow U(a,b)$ means x is a realization of a random variable uniformly distributed in $[a,b]$. }

$$x_{i,k} = A_i x_{i,k-1} + B_i(u_{i,k-1} + w_{i,k-1})$$

$$x_{j,k} = A_j x_{j,k-1} + B_j(u_{j,k-1} + w_{j,k-1})$$

    if $y_{i,k} - x_{i,k} \in [\![\mathbf{v}_{i,k}]\!]$ && $y_{j,k} - x_{j,k} \in [\![\mathbf{v}_{j,k}]\!]$ && $z_{i,j,k} - \|p_{i,k} - p_{j,k}\|_2 \in [\![\mathbf{r}_{i,j,k}]\!]$ then $x_{i,k}^{(m)} \leftarrow x_{i,k}$.

    end if

      end for

    end for

    The posterior estimated set of the position of UAV i is given by

$$a_{i,x,k} = \min\{p_{i,x,k}^{(m)}\}, b_{i,x,k} = \max\{p_{i,x,k}^{(m)}\};$$

$$a_{i,y,k} = \min\{p_{i,y,k}^{(m)}\}, b_{i,y,k} = \max\{p_{i,y,k}^{(m)}\}.$$

    end for

  $k = k + 1$;

end loop

$$\xi_1 \in \left[0, \frac{\overline{R}_{i,j,k}}{\cos\left(\frac{\pi q}{M}\right)}\right] \times \left[0, \frac{\overline{R}_{i,j,k}}{\cos\left(\frac{\pi q}{M}\right)}\right]\Bigg\}. \tag{35}$$

Notice that point $A = p_{j,k}^0 + \underline{R}_{i,j,k} \begin{bmatrix} \cos\left(\frac{2\pi(q+1)}{M}\right) \\ \sin\left(\frac{2\pi(q+1)}{M}\right) \end{bmatrix}$ and the middle point of CD can be represented by $p_{j,k}^0 + \overline{R}_{i,j,k} \begin{bmatrix} \cos\left(\frac{\pi(2q+1)}{M}\right) \\ \sin\left(\frac{\pi(2q+1)}{M}\right) \end{bmatrix}$, $\sigma_{r,q,1}$ and $\sigma_{r,q,2}$ can be obtained by $A \in \{z : H_{r,q} z = \sigma_{r,q,1}\}$ and $C \in \{z : H_{r,q} z = \sigma_{r,q,2}\}$.

Write $[\sigma_{r,q,1}, \sigma_{r,q,2}]$ as

$$\mathcal{Z}_\sigma = \{\sigma = (\sigma_{r,q,2} - \sigma_{r,q,1})\xi_2 + \sigma_{r,q,1},\ \xi_2 \in [0,1]\}. \tag{36}$$

Thus, (17) can be obtained by $\mathcal{Z}_{i,j,k}^{r,q} = \mathcal{Z}_{COD} \cap_{H_{r,q}} \mathcal{Z}_\sigma$ with Lemma 2. ■

Appendix E
Proof of Theorem 2

Proof: For a fixed point $p_{j,k}^* \in \mathcal{Z}_{j,k}^{abs}$, the following two inequalities hold for $\forall p_{i,k} \in \mathcal{Z}_{i,k}^{abs}$,

$$\begin{aligned} \|p_{j,k}^* - p_{i,k}\|_2 &\le \|p_{j,k}^* - p_{j,k}\|_2 + \|p_{j,k} - p_{i,k}\|_2 \\ &\le d(\mathcal{Z}_{j,k}^{abs}) + \overline{R}_{i,j,k} = \widetilde{\overline{R}}_{i,j,k}. \end{aligned} \tag{37}$$

$$\begin{aligned} \|p_{j,k}^* - p_{i,k}\|_2 &\ge \|p_{j,k} - p_{i,k}\|_2 - \|p_{j,k}^* - p_{j,k}\|_2 \\ &\ge \underline{R}_{i,j,k} - d(\mathcal{Z}_{j,k}^{abs}) = \widetilde{\underline{R}}_{i,j,k}. \end{aligned} \tag{38}$$

Hence, $p_{i,k} \in \{s : \widetilde{\underline{R}}_{i,j,k} \le \|s\|_2 \le \widetilde{\overline{R}}_{i,j,k}\}$, which implies the proof of Theorem 2 has been transformed into the one-fixed-point scene as in Proposition 3. Thus, the rest of the proof is the same as Proposition 3. ■

Appendix F
Distributed Particle Filter Based SME Algorithm

The particle-filter-based distributed SME algorithm is presented as in Algorithm 3:

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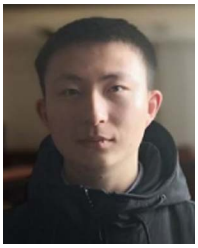
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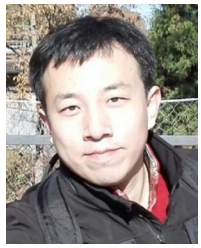
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